

- Direct methods for solving systems of linear equations -

Exercise 1: True or False

Are the following statements true or false?

- (1) The matrix $A = \begin{pmatrix} 2 & 1 \\ 1 & 1 \end{pmatrix}$ admits a Cholesky decomposition.
- (2) The matrix $B = \begin{pmatrix} 1 & -2 & 0 \\ 1 & -1 & 0 \\ 0 & 0 & 3 \end{pmatrix}$ is symmetric positive definite.
- (3) The matrix B above admits an LU decomposition.
- (4) The matrix $\begin{pmatrix} 1 & -1 \\ 1 & 3 \end{pmatrix}$ can be written as $C^t C$.
- (5) The matrix $\begin{pmatrix} 1 & 1 \\ 1 & 5 \end{pmatrix}$ admits a Cholesky decomposition $A = C^t C$ with $C = \begin{pmatrix} -1 & -1 \\ 0 & -2 \end{pmatrix}$.
- (6) Let $A = \begin{pmatrix} 0 & 1 \\ 1 & 0 \\ 1 & 1 \end{pmatrix}$
 - (a) The matrix AA^t admits a Cholesky decomposition.
 - (b) The matrix $A^t A$ admits a Cholesky decomposition.

Exercise 2: Gaussian elimination

We want to solve the linear system $Ax = b$ with

$$A = \begin{pmatrix} 1 & -2 & 0 & 1 \\ -2 & 13 & 3 & 7 \\ 0 & 3 & 2 & 1 \\ 4 & -8 & -2 & 12 \end{pmatrix} \quad \text{and} \quad b = \begin{pmatrix} b_1 \\ b_2 \\ b_3 \\ b_4 \end{pmatrix}$$

- (1) Give the triangular system $A^{(4)} = b^{(4)}$ obtained by Gauss elimination.
- (2) Deduce $\det(A)$ and the inverse matrix A^{-1} .

- (3) Deduce from question (1) the inverse L^{-1} of LU decomposition.

Exercise 3: Gauss elimination pivot

Let's assume numbers are represented in floating-point decimal with 3 significant digits, and the result of operations is rounded to 3 significant digits. Consider the linear system $Ax = b$ with

$$\begin{pmatrix} 10^{-4} & 1 \\ 1 & 1 \end{pmatrix} \text{ and } b = \begin{pmatrix} 1 \\ 2 \end{pmatrix}$$

- (1) Solve using the Gauss elimination method by choosing 10^{-4} as the first pivot.
- (2) Solve using the Gauss elimination method by choosing the second row as the pivot at the first step.
- (3) Conclude.

Exercise 4: About the decomposition LL^t

Let A be a square matrix of order n , symmetric positive definite, and dense. We want to solve the system $Ax = b$.

Two methods are proposed to solve this system:

- Calculate A^2 , perform the LL^t decomposition of A^2 , solve the system $LL^t x = b$.
- Calculate the LL^t decomposition of A , solve the systems $LL^t y = b$ and $LL^t x = y$.

Calculate the number of elementary operations required for each of the two methods and compare.

Exercise 5: LU and Cholesky decompositions

Let $M = \begin{pmatrix} 4 & 2 & 0 \\ 2 & 4 & 1 \\ 0 & 1 & 1 \end{pmatrix}$

- (1) Calculate the principal minors of M . Deduce that M admits LU and Cholesky decompositions.
- (2) Provide the LU decomposition of M .
- (3) Provide the Cholesky decomposition of M .
- (4) Solve the system $Ax = b$ using the previous decomposition such that $b = \begin{pmatrix} 8 \\ 13 \\ 5 \end{pmatrix}$